

EDS Assignment

PS50008A Research Methods | Term 2 Spring 2026

Gordon Wright

21 January 2026

Key Information

Deadline	Friday 20 March 2026, 12 noon
Submission	Via Moodle submission link
Format	Worked statistics assignment
Weight	20%
Pass mark	50%

Task Structure

You will receive a research scenario with **pre-generated R output**. Your task is to interpret the output and demonstrate your understanding of the statistical analysis - no coding or hand calculations required.

Section A: Statistical Mechanics (50 marks)

1. State the researcher's hypothesis (5 marks)
2. Identify design type, IV, DV, data type (16-20 marks)
3. **Interpret R descriptive output** and report in APA format (10 marks)
4. **Interpret R t.test() output** and justify test choice (15 marks)
5. Full APA results paragraph from R output (5 marks)
6. **NEW: Error detection** - identify what's wrong with a given analysis (5 marks)

Section B: Methods and Theory (50 marks)

7. Suggest TWO methodological improvements
 8. Discuss individual differences / extensions
 9. Propose a more ecologically valid alternative (minimum half page)
-

Critical Requirements

⚠ Demonstrate Understanding

You **MUST** demonstrate your understanding - not just get the right answer.

- Full marks = right test AND proper justification
- No punitive marking, so if you get something wrong, you just get zero for that item
- You are expected to use the terminology and concepts we have been discussing in class
- Pay attention to the glossaries and readings provided

What “Demonstrate Understanding” Means

Aspect	What we’re looking for
Test selection	Why this test? Link to design and data type
Hypothesis	Clear statement with direction justified
Reading output	Correctly identify key values (t, df, p, CI)
Interpretation	What does the result mean in context?
Error detection	Identify when an analysis doesn’t match the design
Improvements	Thoughtful, specific, feasible suggestions

Preparation

- Practice interpreting R output from lab exercises (weeks 12-16)
- Review APA reporting format for statistics
- Understand the logic behind test selection
- Be comfortable reading and interpreting R output
- Practice error detection exercises (what’s wrong with this analysis?)

Example R Output You’ll Encounter

Descriptive Statistics

```
> data %>%  
+   group_by(condition) %>%  
+   summarise(Mean = mean(RT), SD = sd(RT), n = n())  
  
# A tibble: 2 x 4
```

	condition	Mean	SD	n
	<chr>	<dbl>	<dbl>	<int>
1	30_degree	285.	42.3	10
2	150_degree	312.	51.1	10

T-Test Output

```
> t.test(data$RT_30, data$RT_150, paired = TRUE)

Paired t-test

data:  data$RT_30 and data$RT_150
t = -2.847, df = 9, p-value = 0.0192
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -49.82 -5.38
sample estimates:
mean of the differences
      -27.6
```

You will NOT need to write R code. Your task is to:

1. Read the output correctly
2. Explain what the numbers mean
3. Report findings in APA format
4. Detect when an analysis contains errors